

IFS506 – Decision Support Systems

A **Decision Support System (DSS)** is a computer-based information system that helps individuals, managers, professionals, and organizations make better decisions by collecting, processing, analyzing, and presenting data in a useful form.

A DSS supports decision-making in several important ways, including:

- Relevant Data
- Analytical Models
- Visual Reports
- Forecasts
- Simulations
- Recommendations

Purpose of DSS

The main purpose of DSS is to improve the quality, speed, accuracy, and confidence of decision-making.

Other key purposes of a DSS include:

- Improving Decision Quality
- Supporting Problem Solving:
- Enhancing Speed of Decision-Making
- Providing Data Analysis and Insights
- Facilitating Forecasting and Planning
- Supporting Strategic, Tactical, and Operational Decisions
- Reducing Uncertainty and Risk
- Improving Organizational Efficiency:
- Supporting Collaboration and Communication
- Assisting Human Judgment Rather Than Replacing It

EXAMPLES OF DSS

Examples of Decision Support Systems (DSS) in different fields include:

Healthcare

- Clinical Decision Support Systems (CDSS)
- Disease diagnosis systems
- Patient monitoring systems
- Drug interaction alert systems

Business and Finance

- Financial planning systems
- Credit scoring systems
- Investment analysis systems
- Sales forecasting systems

Education

- Student performance analysis systems
- Admission decision systems
- Academic recommendation systems

Agriculture

- Crop management systems
- Weather forecasting systems
- Irrigation planning systems

Transportation and Logistics

- Traffic management systems
- Route optimization systems
- Fleet management systems

Manufacturing

- Production planning systems
- Supply chain management systems
- Inventory control systems

Government and Public Sector

- Disaster management systems
- Urban planning systems
- Crime analysis systems

E-commerce and Marketing

- Recommendation systems (e.g., Amazon, Netflix)
- Customer relationship management (CRM) systems
- Market trend analysis systems

Environmental Management

- Climate monitoring systems
- Environmental risk assessment systems
- Pollution monitoring systems

Military and Security

- Battlefield decision systems
- Surveillance and threat detection systems
- Strategic planning systems

Some well-known real-world DSS examples include:

- IBM Watson Health
- SAP Business Intelligence
- Oracle Decision Support Systems
- Google Maps route optimization
- Amazon recommendation engine
- Netflix recommendation system

NATURE OF DECISION-MAKING

What is Decision-Making?

Decision-making is the process of selecting the best option from several alternatives.

Example: Personal Decision-Making

Problem: A student needs to choose a project topic.

Alternatives:

- Malaria Prediction using Deep Learning
- Student Performance Prediction
- Crop Yield Forecasting

Decision Process:

- Evaluate data availability.
- Consider personal interest.
- Assess feasibility within the project timeline.

Decision:

The student selects **Malaria Prediction using Deep Learning** because relevant datasets are available and the topic aligns with their research interests.

Example: Business Decision-Making

Problem: A company wants to increase sales.

Alternatives:

- Increase advertising budget.
- Reduce product prices.
- Introduce a new product.

Decision Process:

- Analyze market trends.
- Estimate costs and expected returns.
- Evaluate risks.

Decision:

The company decides to **increase digital advertising** because it offers the highest projected return on investment.

Key Elements of Decision-Making

1. Identifying the Problem or Objective
2. Collecting Relevant Information
3. Generating Possible Alternatives
4. Evaluating the Alternatives
5. Selecting the Best Option
6. Implementing the Decision
7. Monitoring and Reviewing the Outcome

Types of Decisions**1. Structured Decisions**

These are routine and repetitive decisions with clear rules.

2. Semi-Structured Decisions

These involve both clear rules and human judgment.

3. Unstructured Decisions

These are complex decisions with no fixed procedure.

Feature	Structured Problems	Semi-Structured Problems	Unstructured Problems
Meaning	Problems with clear procedures and well-defined solutions.	Problems that are partly clear and partly require human judgment.	Problems with no clear procedures or definite solutions.
Decision-Making Process	Fully programmable and routine.	Partly programmable and partly judgment-based.	Mainly based on intuition, experience, and creativity.
Level of Uncertainty	Low uncertainty.	Moderate uncertainty.	High uncertainty.
Use of Rules and Procedures	Standard rules and procedures exist.	Some rules exist, but not enough for complete decisions.	No standard rules or procedures available.
Human Judgment Required	Very little.	Moderate.	Very high.
Ease of Automation	Easily automated by computers.	Partially automated.	Difficult to automate.
Nature of Decision	Routine and repetitive.	Combination of routine and non-routine.	Non-routine and unique.
Examples	Payroll processing, inventory calculation, ATM transactions.	Loan approval, medical diagnosis, budgeting decisions.	Business strategy, crisis management, policy formulation.

DSS is especially useful for **semi-structured and unstructured decisions**.

Components of a Decision Support System

Components of a Decision Support System (DSS) are the essential parts that work together to help users analyze information, evaluate options, and make better decisions.

- 1) Data Management Component
- 2) Model Management Component
- 3) Knowledge Management Component
- 4) User Interface (Dialogue Management Component)
- 5) Users (Decision Makers)

Types of Decision Support Systems

Data-Driven DSS

This focuses on accessing and analyzing large volumes of data.

Example

- sales dashboards

Model-Driven DSS

This uses mathematical and analytical models.

Example

- financial forecasting systems

Knowledge-Driven DSS

This uses expert knowledge and rules to support decisions.

Example

- medical diagnosis support systems

Document-Driven DSS

This helps users search, analyze, and manage documents.

Example

- legal case search systems
- policy document analysis systems
- academic research support tools

Communication-Driven DSS

This supports group decision-making and collaboration.

Example

- online meeting systems

Data Visualization in DSS

Data visualization is the graphical representation of data using charts, graphs, maps, diagrams, and interactive visuals.

Importance of Data Visualization in DSS

Data visualization is important because it:

- simplifies complex data
- reveals hidden patterns
- supports faster decision-making
- improves communication
- helps detect errors and anomalies
- makes reports easier to understand

Example

A sales manager can understand monthly sales performance faster from a line graph than from a long spreadsheet.

Data Representation

Data representation refers to the way data is organized, encoded, and presented for analysis and interpretation.

In DSS, data can be represented in different forms, including:

- tables
- charts
- graphs
- maps
- trees
- networks
- images
- 3D models

Common Forms of Data Representation

1. Tables

A table is a method of organizing data into rows and columns. Each row represents a single record or item, while each column represents a specific attribute or category of information.

Best Used For

- Displaying exact values
- Storing structured records
- Comparing specific entries across different records

Example

Product	Units Sold	Unit Price (₦)	Total Sales (₦)
Rice	150	45,000	6,750,000
Beans	100	60,000	6,000,000
Garri	200	25,000	5,000,000

2. Bar Charts

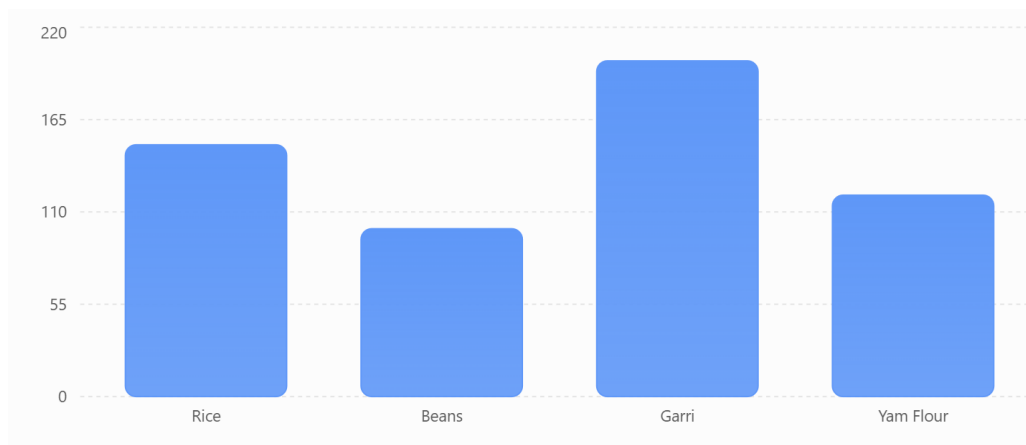
A bar chart is a graphical method of representing data using rectangular bars of equal width. The length or height of each bar represents the value of a particular category.

Best Used For

- Comparing quantities across categories
- Showing differences between groups
- Identifying the highest and lowest values

Example:

Comparison of monthly sales across product categories.



3. Line Graphs

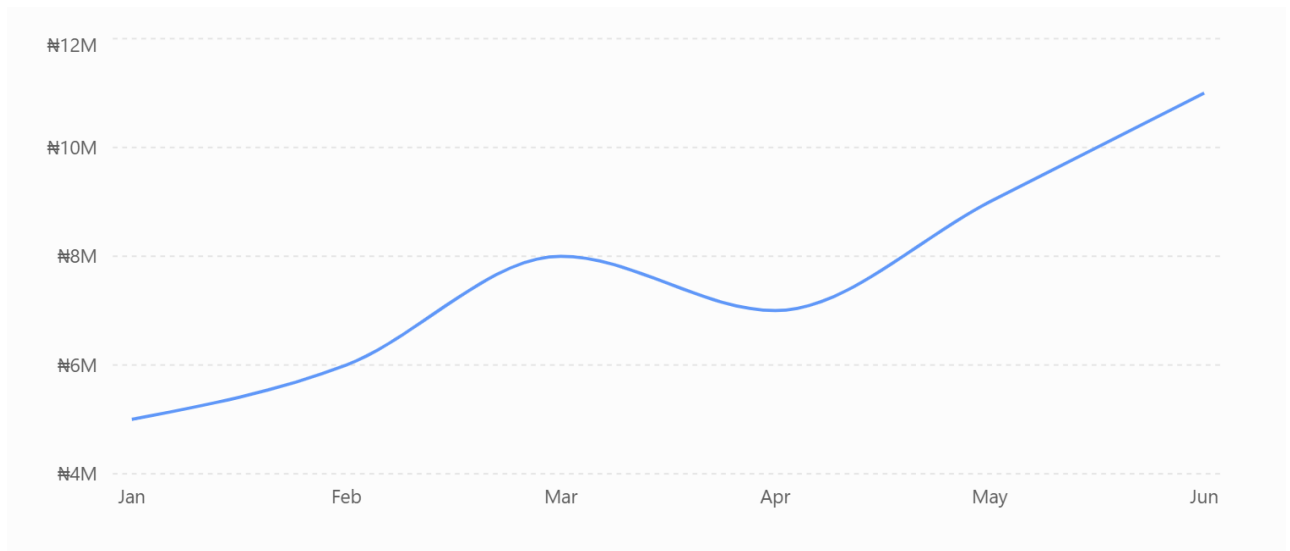
A line graph is a graphical representation of data that uses points connected by straight lines to show how values change over time. Each point on the graph represents a data value at a specific time, and the connecting lines make it easy to observe trends, increases, decreases, or fluctuations.

Best Used For

- Showing changes over time
- Identifying trends and patterns
- Monitoring performance over a period

Example

Revenue generated over six consecutive months.



4. Pie Charts

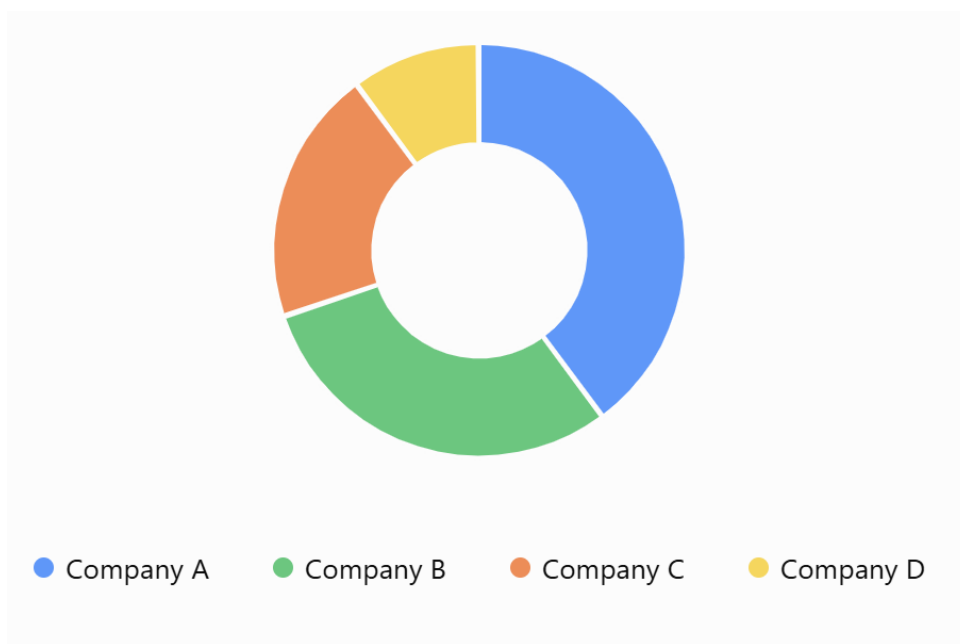
A pie chart is a circular chart that is divided into slices, where each slice represents a portion or percentage of the whole. The size of each slice is proportional to the value it represents.

Best Used For

- Showing proportions or percentages of a whole
- Displaying the composition of a dataset
- Comparing the contribution of different categories to a total

Example

Percentage contribution of each company to the total market.



5. Scatter Plots

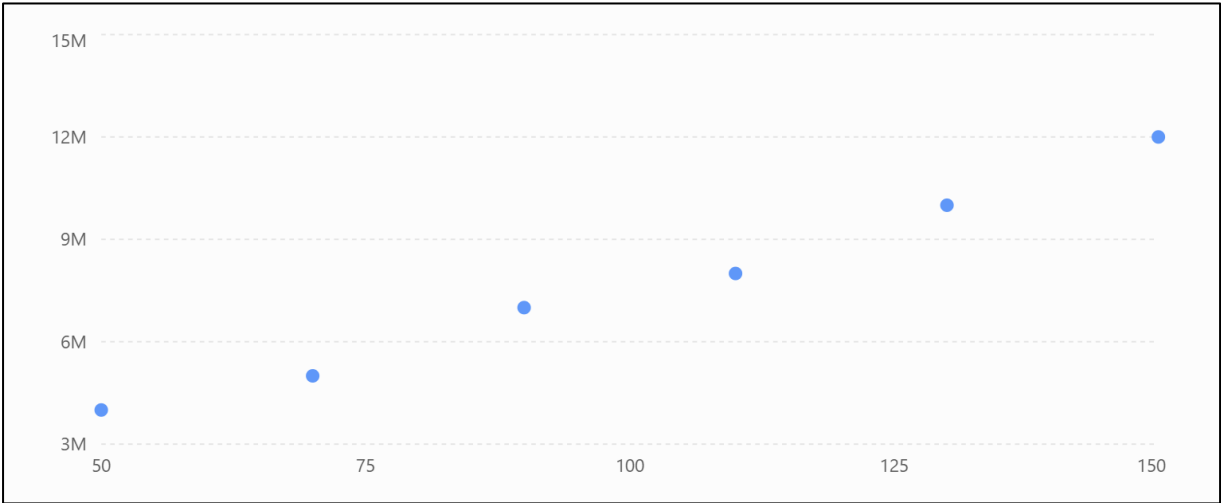
A scatter plot is a graph that uses dots to represent the relationship between two numerical variables. Each dot represents one observation, with one variable plotted on the horizontal axis (X-axis) and the other on the vertical axis (Y-axis).

Best Used For

- Showing the relationship between two variables
- Identifying correlations and patterns
- Detecting outliers (unusual data points)

Example

Relationship between advertising expenditure and monthly sales revenue.



6. Heat Maps

A heat map is a graphical representation of data that uses different colours or colour intensities to show the magnitude of values.

Best Used For

- Showing data intensity across locations or categories
- Identifying patterns, hotspots, and trends
- Highlighting areas that require attention

Example

Daily Sales by Store Location

Store Location	Sales Level	Heat Level
Akure	₦5,000,000	High
Lagos	₦4,200,000	Medium-High
Ibadan	₦3,000,000	Medium
Abuja	₦2,000,000	Low
Benin	₦1,500,000	Low

7. Maps / Geographic Visualization

Maps or Geographic Visualizations are graphical representations of data based on geographical locations. They display information on a map so that users can understand where events or activities occur rather than just how many occur.

Best Used For

- Displaying data based on geographic locations
- Identifying spatial patterns and trends
- Supporting location-based decision-making

Example

Location	Number of Sales
Akure	250
Lagos	500
Ibadan	320
Abuja	410
Benin	180



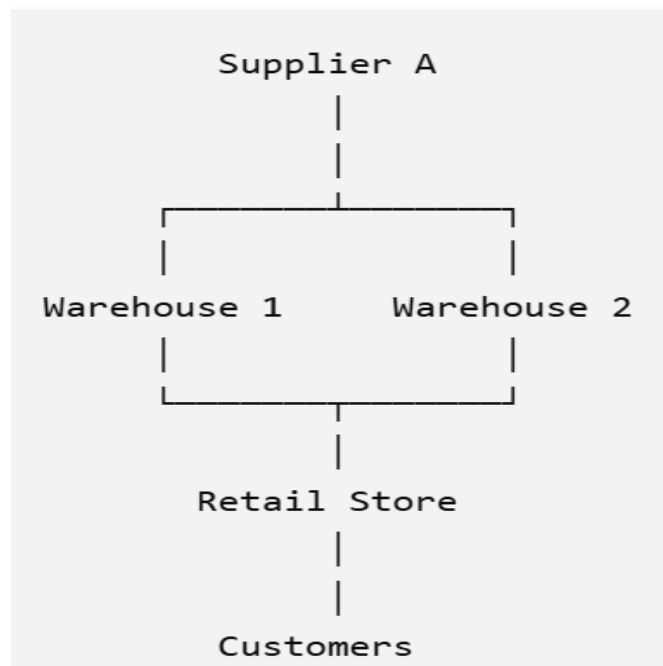
8. Network Diagrams

A network diagram is a graphical representation that shows the relationships among different entities (nodes). The entities are represented as nodes (circles or boxes), while the relationships between them are represented by lines (links or edges).

Best Used For

- Showing relationships among entities
- Visualizing communication or information flow
- Identifying connections and dependencies

Example



Visualization Toolkits

Visualization toolkits are software libraries, frameworks, or platforms used to create graphical and interactive representations of data.

They help developers and analysts build dashboards, charts, maps, and advanced visual displays.

Examples of Visualization Toolkits

1. Microsoft Power BI

Microsoft Power BI is a business intelligence (BI) and data visualization tool developed by Microsoft. It enables users to connect to multiple data sources, analyze data, and create interactive dashboards, reports, and visualizations that support decision-making.

Main Features

- Interactive dashboards
- Charts, graphs, maps, and tables
- Real-time data monitoring
- Data analysis and reporting
- Integration with Excel, SQL Server, and cloud services

Best Used For

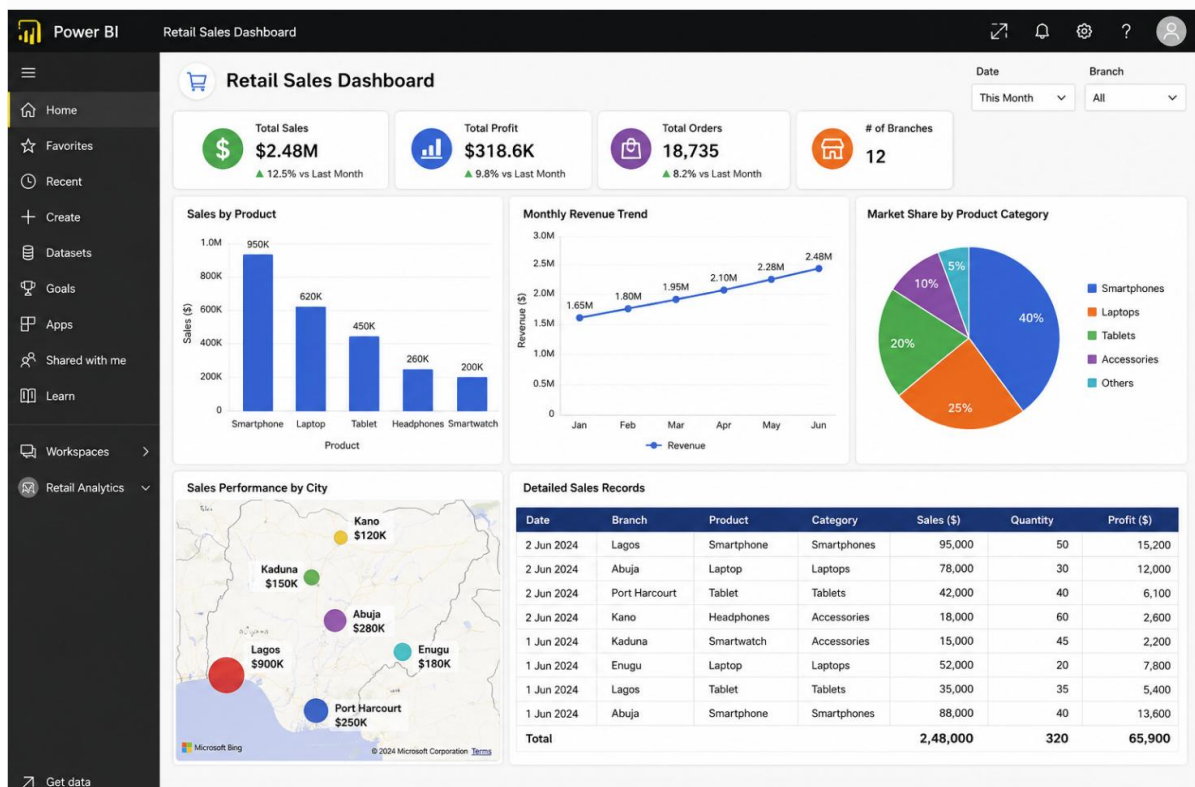
- Business intelligence dashboards
- Sales and financial reports
- Performance monitoring
- Data analysis and decision-making

Example

A retail company uses **Power BI** to monitor monthly sales across its branches.

The dashboard may include:

- A **bar chart** showing sales by product.
- A **line graph** showing monthly revenue trends.
- A **pie chart** showing market share by product category.
- A **map** showing sales performance across different cities.
- A **table** displaying detailed sales records.



Managers can view all these visualizations on a single dashboard and quickly identify trends, top-performing products, and underperforming branches.

Advantages of Microsoft Power BI

- Easy to create interactive dashboards.
- Connects to multiple data sources.
- Supports real-time data updates.
- Provides a wide variety of visualization options.
- Enables users to share reports online.

Limitations of Microsoft Power BI

- Advanced features require training.
- Large datasets may require premium licenses.
- Performance may slow when handling extremely large volumes of data.

2. Tableau

Tableau is a data visualization and business intelligence (BI) platform that helps users transform raw data into interactive charts, graphs, maps, and dashboards. It allows users to analyze large amounts of data and discover trends without requiring advanced programming skills.

Main Features

- Interactive dashboards
- Drag-and-drop interface
- Charts, graphs, maps, and tables
- Real-time data visualization
- Integration with multiple data sources (Excel, SQL, Oracle, cloud databases, etc.)

Best Used For

- Business intelligence and analytics
- Sales and marketing analysis
- Financial reporting
- Customer behavior analysis
- Executive dashboards

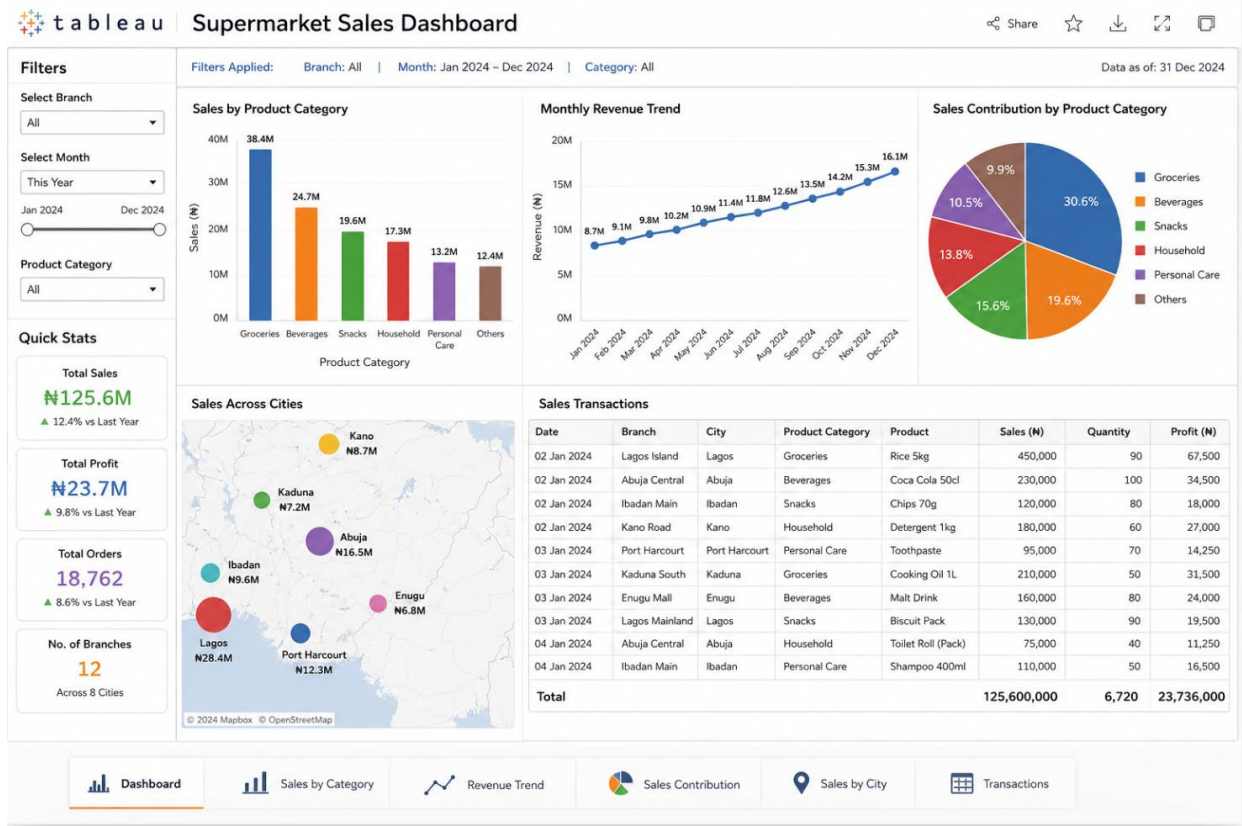
Example

A supermarket chain uses **Tableau** to analyze sales performance across its branches.

The dashboard may include:

- A **bar chart** showing sales by product category.
- A **line graph** showing monthly revenue trends.
- A **pie chart** showing the percentage contribution of each product category.

- A **map** displaying sales across different cities.
- A **table** containing detailed sales transactions.
- **Interactive filters** that allow managers to view sales by branch, month, or product category.



Managers can click on any chart or filter to explore the data further and make informed business decisions.

Advantages of Tableau

- Easy-to-use drag-and-drop interface.
- Creates highly interactive dashboards.
- Connects to many data sources.
- Processes and visualizes large datasets efficiently.
- Supports real-time data analysis.

Limitations of Tableau

- Requires a paid license for full functionality.
- Advanced analysis may require additional training.
- Can consume significant computing resources when handling very large datasets.

3. Matplotlib

Matplotlib is an open-source Python library used for creating a wide variety of charts, graphs, and data visualizations. It is one of the most popular visualization libraries in Python and is widely used in data science, engineering, research, and business analytics.

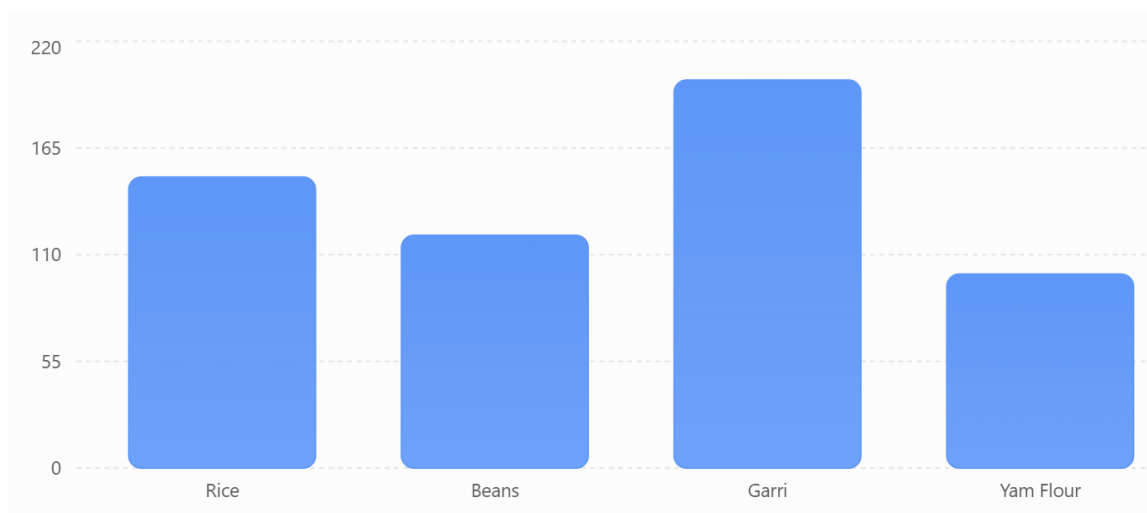
Main Features

- Creates bar charts, line graphs, pie charts, scatter plots, histograms, and more.
- Highly customizable charts.
- Integrates with libraries such as **NumPy**, **Pandas**, and **SciPy**.
- Suitable for scientific and business data visualization.
- Can save charts as images (PNG, JPG, PDF, SVG).

Best Used For

- Data analysis using Python
- Scientific and engineering visualization
- Statistical analysis
- Machine learning projects
- Academic research

Example of a bar chart generated using Matplotlib.



Advantages of Matplotlib

- Free and open source.
- Produces high-quality charts.
- Highly customizable.
- Integrates well with Python data analysis libraries.
- Suitable for research and scientific computing.

Limitations of Matplotlib

- Requires knowledge of Python programming.
- Interactive dashboards are limited compared to Power BI and Tableau.
- Complex visualizations may require more code.

4. Seaborn

Seaborn is an open-source Python data visualization library built on top of Matplotlib. It provides a simpler way to create attractive, informative, and statistically meaningful charts with less code than Matplotlib.

Main Features

- Creates high-quality statistical graphics.
- Built on top of Matplotlib.
- Integrates seamlessly with Pandas DataFrames.
- Provides attractive default styles and color themes.
- Supports statistical plots such as heat maps, box plots, violin plots, pair plots, and regression plots.

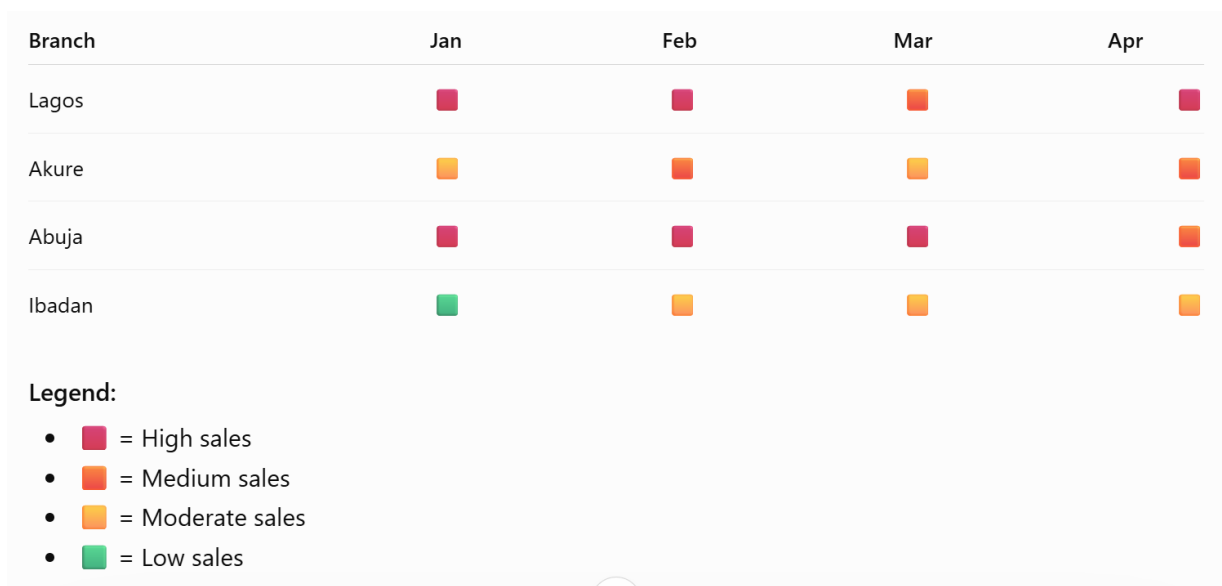
Best Used For

- Statistical data visualization
- Exploratory Data Analysis (EDA)
- Machine learning projects
- Business analytics
- Research and academic studies

Example

A supermarket uses **Seaborn** to analyze customer purchasing patterns.

Below is an example of a **heat map** that could be created using Seaborn.



Advantages of Seaborn

- Easy to use with fewer lines of code than Matplotlib.
- Produces attractive and professional-looking charts.
- Excellent for statistical visualization.

- Integrates well with Pandas and NumPy.
- Provides built-in themes and color palettes.

Limitations of Seaborn

- Requires knowledge of Python programming.
- Less flexible than Matplotlib for highly customized visualizations.
- Some advanced chart customization still requires Matplotlib.

5. Plotly

Plotly is an open-source data visualization library used to create interactive charts, graphs, maps, and dashboards. It supports Python, R, JavaScript, and other programming languages, making it popular in data science, business analytics, and web applications.

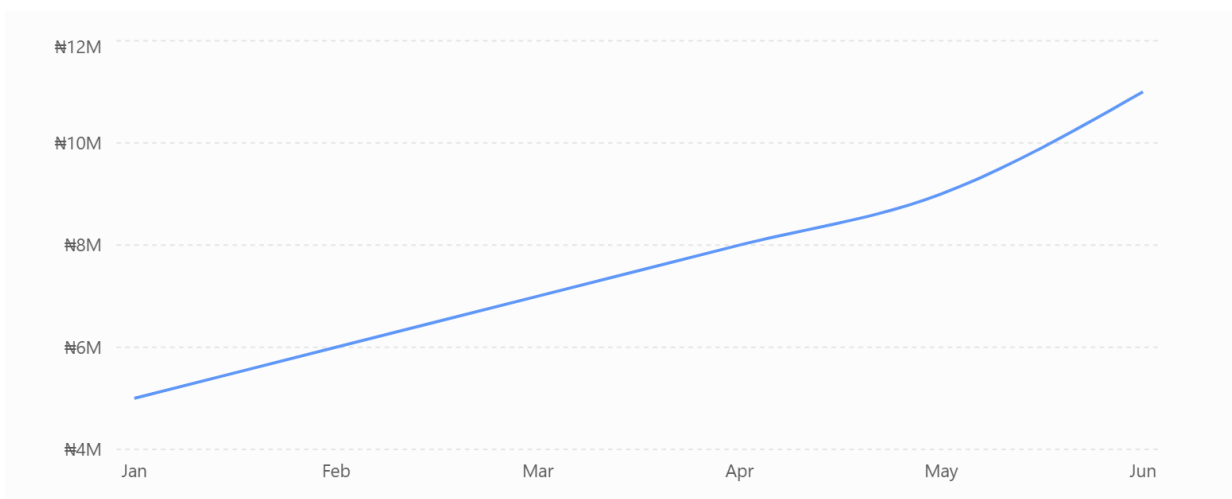
Main Features

- Creates interactive charts and dashboards.
- Supports bar charts, line graphs, pie charts, scatter plots, maps, and 3D charts.
- Allows users to zoom, pan, and hover over data.
- Integrates with Python libraries such as Pandas and NumPy.
- Can be used with Dash to build web-based analytical dashboards.

Best Used For

- Interactive business dashboards
- Data science and machine learning
- Financial analysis
- Scientific visualization
- Web-based data analytics applications

Example of an interactive line chart created using Plotly.



Advantages of Plotly

- Produces highly interactive and attractive visualizations.
- Supports a wide range of chart types, including 3D charts and maps.
- Integrates well with Python and web applications.
- Suitable for creating business intelligence dashboards.
- Allows users to explore data through zooming, hovering, and filtering.

Limitations of Plotly

- Requires knowledge of programming (especially Python or JavaScript).
- More advanced dashboards may require additional libraries such as Dash.
- Can consume more computing resources than simpler visualization libraries.

6. D3.js

D3.js (Data-Driven Documents) is an open-source JavaScript library used to create highly customized, interactive, and web-based data visualizations. D3.js is widely used to build interactive dashboards, infographics, maps, and data-driven web applications.

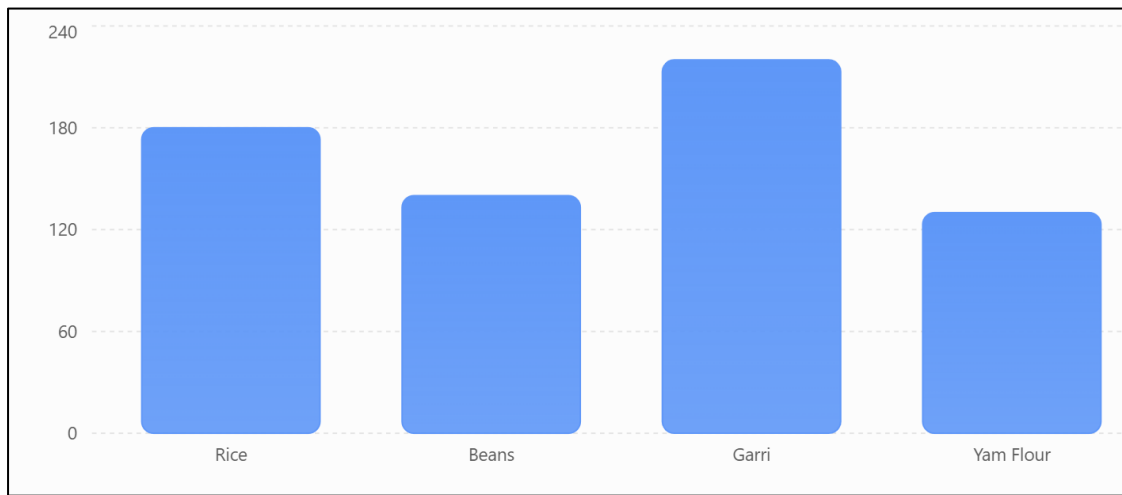
Main Features

- Creates highly interactive and customizable charts.
- Supports bar charts, line graphs, pie charts, scatter plots, maps, network diagrams, and tree diagrams.
- Uses HTML, CSS, SVG, and JavaScript for visualization.
- Allows animations, transitions, zooming, and filtering.
- Suitable for building web-based dashboards.

Best Used For

- Interactive web applications
- Business dashboards
- Scientific and research visualization
- Geographic information systems (GIS)
- Network and social media analysis

Example of a web-based interactive chart created using D3.js.



Advantages of D3.js

- Highly flexible and customizable.
- Produces rich, interactive visualizations.
- Supports animations and dynamic updates.
- Ideal for web-based dashboards and applications.
- Can visualize complex datasets and relationships.

Limitations of D3.js

- Requires knowledge of JavaScript, HTML, CSS, and SVG.
- Has a steeper learning curve than Power BI or Tableau.
- Developing complex visualizations can take more time.

7. Google Charts

Google Charts is a free JavaScript library developed by Google for creating simple, interactive, and web-based charts. Google Charts is easy to use and is suitable for websites and small business dashboards that require interactive visualizations without extensive programming.

Main Features

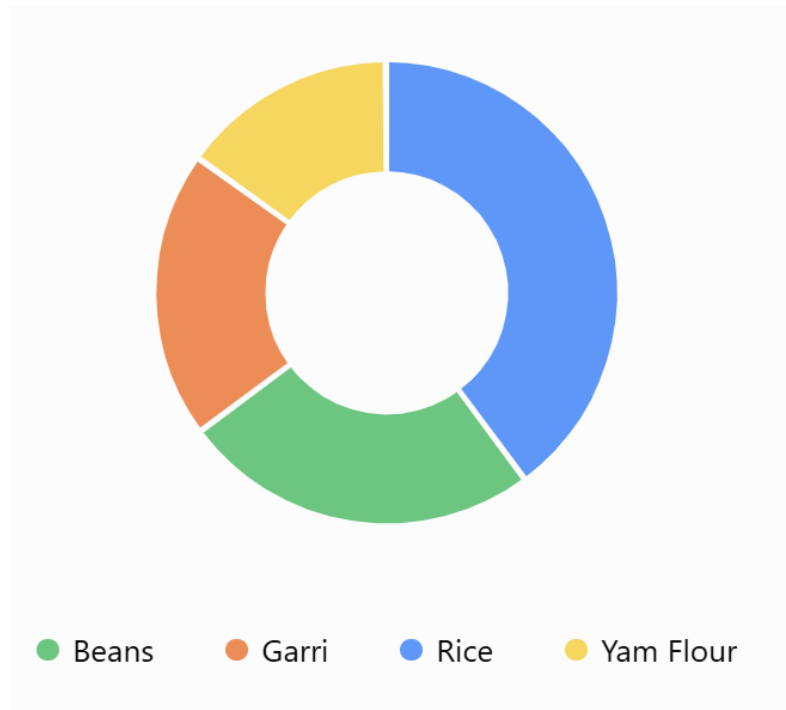
- Creates interactive web-based charts.
- Supports bar charts, line graphs, pie charts, scatter plots, maps, and tables.
- Easy to integrate into websites.
- Allows users to hover over chart elements to view detailed information.
- Works well with data from spreadsheets and databases.

Best Used For

- Simple web-based dashboards
- Business reports
- Educational websites

- Sales and financial analysis
- Interactive data presentation

Example of a pie chart created using Google Charts.



Advantages of Google Charts

- Free and easy to use.
- Produces interactive charts with minimal code.
- Supports many common chart types.
- Works well in web browsers.
- Integrates easily with web applications and databases.

Limitations of Google Charts

- Offers fewer customization options than D3.js.
- Requires an internet connection for some features.
- Less suitable for highly complex visualizations.

Choosing a Visualization Toolkit

When selecting a visualization toolkit, several factors should be considered to ensure it meets the needs of the users and the organization.

1. Type of Data
2. Level of Interactivity Required
3. User Skill Level
4. Cost

5. Deployment Platform
6. Integration with Databases
7. Performance Requirements

Difference between Data Representation and Visualization Toolkits

The key difference is that Data Representation refers to what you use to display data, while Visualization Toolkits refer to the software or libraries you use to create those displays.

Data Representation	Visualization Toolkits
Refers to the format or method used to present data.	Refers to the software, applications, or programming libraries used to create visual representations of data.
Answers the question: "How is the data displayed?"	Answers the question: "What tool is used to create the display?"
Includes tables, bar charts, line graphs, pie charts, scatter plots, heat maps, maps, and network diagrams.	Includes Microsoft Power BI, Tableau, Matplotlib, Seaborn, Plotly, D3.js, and Google Charts.
Focuses on the type of visualization.	Focuses on the technology used to produce the visualization.
It is a concept or method.	It is a software tool or programming library.

Scientific Visualization

Scientific visualization is the visual representation of scientific data, usually from experiments, simulations, or measurements.

Examples

Scientific visualization is used in:

- climate modeling
- physics simulations
- chemistry modeling
- astronomy

Importance

Scientific visualization helps to:

- interpret large scientific datasets
- observe patterns
- test hypotheses

- communicate research findings
- identify anomalies

Medical Visualization

Medical visualization is the use of graphical and imaging techniques to represent medical data for diagnosis, treatment planning, and research.

Examples of Medical Visualization

- X-ray images
- CT scans
- MRI scans

Importance in Decision Support

Medical visualization supports:

- diagnosis of diseases
- surgical planning
- treatment monitoring
- medical education
- patient communication

Information Visualization

Information visualization focuses on visualizing abstract data that does not have a natural physical form.

Examples include:

- customer data
- financial records
- social networks

Techniques

Common techniques include:

- dashboards
- charts
- tree maps
- network graphs

Importance

Information visualization helps decision-makers:

- understand business trends
- compare alternatives
- identify relationships
- monitor key performance indicators

Flow Visualization

Flow visualization is the visual representation of movement or flow of substances, objects, or data.

Examples

- airflow around an aircraft
- blood flow in arteries
- traffic flow in a city

Techniques

- arrows
- streamlines
- vector fields
- particle tracing
- animation

Importance in DSS

Flow visualization helps in:

- engineering design
- transport planning
- medical analysis
- network monitoring
- disaster management

Volume Rendering Techniques

Volume rendering is a technique used to display 3D data, especially data that exists inside a volume rather than only on a surface.

It is widely used in medical imaging, scientific visualization, and engineering.

Examples of Volume Data

- CT scan slices
- MRI scan data

- weather data

Common Volume Rendering Techniques

1. Direct Volume Rendering

This displays the entire volume by assigning color and transparency to different data values.

2. Surface Rendering

This extracts surfaces from volume data and displays them as 3D shapes.

3. Ray Casting

Ray casting sends virtual rays through the volume and calculates how light interacts with the data.

It produces high-quality 3D images but may require high computing power.

4. Maximum Intensity Projection

This shows the brightest or highest-intensity values along a viewing direction.

It is often used in medical imaging.

Importance

Volume rendering helps professionals:

- inspect internal structures
- detect abnormalities
- understand spatial relationships
- support complex decision-making

Human Perception in Data Visualization

Human perception refers to how people see, interpret, and understand visual information.

Why Human Perception Matters

Visualization is only useful if users can understand it correctly.

Poor visual design can cause:

- confusion
- wrong interpretation
- poor decisions
- cognitive overload

Principles of Human Perception

1. Color Perception

Colors can highlight important information, but they must be used carefully.

Good practice:

- use consistent colors
- avoid too many colors
- consider color-blind users

2. Size and Position

Humans easily compare size and position.

Bar charts and line charts are often easier to read than complex 3D charts.

3. Pattern Recognition

Humans are good at recognizing patterns.

Visualization should help users see:

- trends
- clusters
- outliers
- gaps

4. Cognitive Load

Cognitive load is the amount of mental effort required to understand information.

Good visualizations reduce cognitive load by being simple and clear.

5. Gestalt Principles

Gestalt principles explain how humans' group visual elements.

Important principles include:

- proximity: nearby items are seen as related
- similarity: similar items are seen as related
- continuity: people follow smooth visual paths
- closure: people mentally complete missing parts

Design of Decision Support Systems

DSS design is the process of planning the structure, components, data flow, models, and user interface of a decision support system.

Steps in DSS Design

Step 1: Identify the Decision Problem

Step 2: Identify Users and Stakeholders

Step 3: Identify Data Requirements

Step 4: Select Analytical Models

Step 5: Design User Interface

Step 6: Test the System

Step 7: Deploy and Monitor

Deployment of Decision Support Systems

Deployment is the process of making a DSS available for real users in an organization.

Deployment Approaches

1. Desktop Deployment
2. Web-Based Deployment
3. Mobile Deployment
4. Cloud-Based Deployment

Deployment Challenges

- user resistance
- poor data quality
- lack of training
- security risks
- integration problems
- high implementation cost

Intelligent Systems in DSS

An intelligent system is a computer system that uses artificial intelligence techniques to learn, reason, adapt, or make recommendations.

In DSS, intelligent systems improve decision-making by analyzing complex data and suggesting suitable actions.

Technologies Used in Intelligent DSS

1. Artificial Intelligence
2. Machine Learning
3. Expert Systems

4. Natural Language Processing

5. Neural Networks

Benefits of Decision Support Systems

DSS provides many benefits, including:

- improved decision quality
- faster analysis
- better use of data
- improved forecasting
- reduced uncertainty
- support for complex decisions
- better communication of insights
- increased organizational efficiency

Limitations of DSS

DSS also has limitations:

- depends on quality of data
- may be expensive to implement
- users may over-rely on system outputs
- models may be inaccurate
- requires technical skills
- may face security and privacy issues

A DSS should support human judgment, not replace it completely.